

has. If the benchmark is risk adjusted, then academics like to call tracking error “*idiosyncratic volatility*” (see sections 3 and 4, below).

Finally, the information ratio is the ratio of alpha to tracking error:

$$\text{Information Ratio} = IR = \frac{\alpha}{\bar{\sigma}}. \quad (10.4)$$

Alpha by itself could be produced by a manager taking large amounts of risk. The information ratio divides the alpha by the risk taken, so it is *the average excess return per unit of risk*. Information ratios above one are not common—although many hedge funds trying to raise money claim to have them. (Since the financial crisis, information ratios on many funds and strategies have come down substantially.)

A special case of equation (10.4) is when the benchmark is the risk-free rate, r_t^f . (Note that the risk-free rate is known at the beginning of the period and applies from $t-1$ to t .) Then, the alpha is the average return in excess of the risk-free rate, $\alpha = \overline{r_t - r_{ft}^f}$ (where the upper bar denotes the sample average of the excess return), and the information ratio coincides with the Sharpe ratio (see chapter 2),

$$\text{Sharpe Ratio} = SR = \frac{\overline{r_t - r_{ft}^f}}{\sigma},$$

where σ is the volatility of the asset.²

2.2 BENCHMARKS MATTER

Martingale’s low volatility strategy is based on the Russell 1000 universe of large stocks, so naturally Jacques takes the Russell 1000 as his benchmark. This is also a convenient benchmark for the asset owner because large investors can buy Russell 1000 index funds for less than 10 basis points per year. Jacques is running an active strategy with relatively high fees and his volatility strategy must offer compelling returns (higher returns, lower risk, or both) relative to the Russell 1000 to attract investors.

A combination of assets or asset classes can also serve as a benchmark. One of Scott and Herscovici’s concerns is that Jacques’s low volatility strategy has a high tracking error, of 6.16%, relative to the Russell 1000. The tracking error is high because Martingale’s product has a low beta of 0.73 relative to the same index. Beta is measured by regressing excess returns of the fund (using T-bills as the risk-free asset) on excess returns of the Russell 1000 (which is the regression implied by the CAPM; see chapter 6):

$$r_t - r_t^f = 0.0344 + 0.7272(r_t^{R1000} - r_t^f) + \varepsilon_t,$$

²In practice, volatilities of excess returns or raw returns are generally almost the same—unless risk-free rates are very volatile, which happens in some emerging markets.